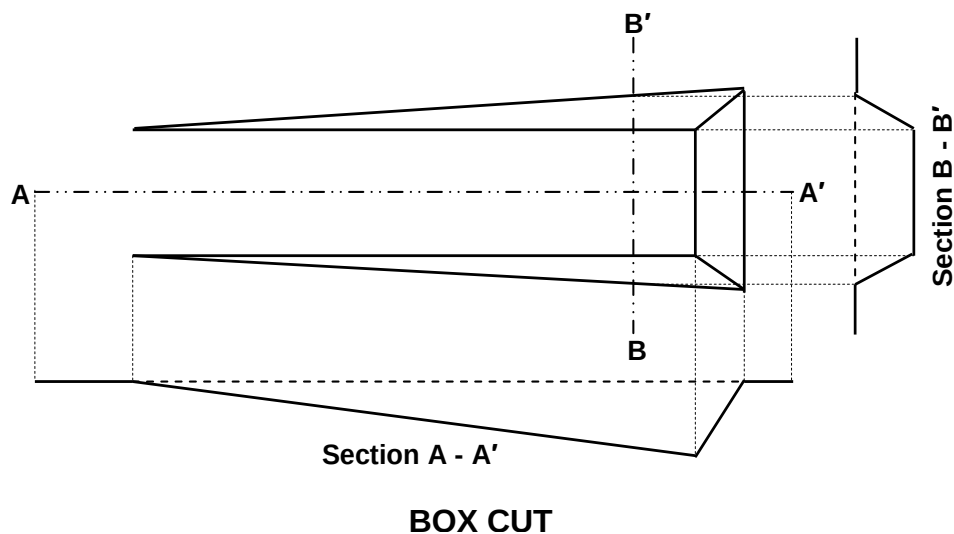


2. OPENING UP OF DEPOSITS

For the physical development of a surface mine it is necessary to establish the different benches in waste/overburden rocks and in ore body/seam(s) for their subsequent handling/extraction. For this an access is to be established first to facilitate the developments of different benches. The access to a surface mine is made through the box cut.

2.1. Box Cut

Box cut is the initial/first cut given for the physical development of a mine. The name 'Box cut' has been attributed to this cut as this cut generally looks like a top open box having an inclined floor and walls on three sides (front wall and two side walls) in case of opening up of subsurface deposits. However, in today's surface mining terminology the initial cut given for the physical development of a mine is called a box cut whether it looks like an open box or not.



2.1.1. Objectives and Types of Box Cut

Objectives

The main objectives of box cut are:

- ❖ To reach the ore body / coal seam
- ❖ To provide a smooth entry to the pit
- ❖ To provide space for development of working / production benches

Types and Applicability

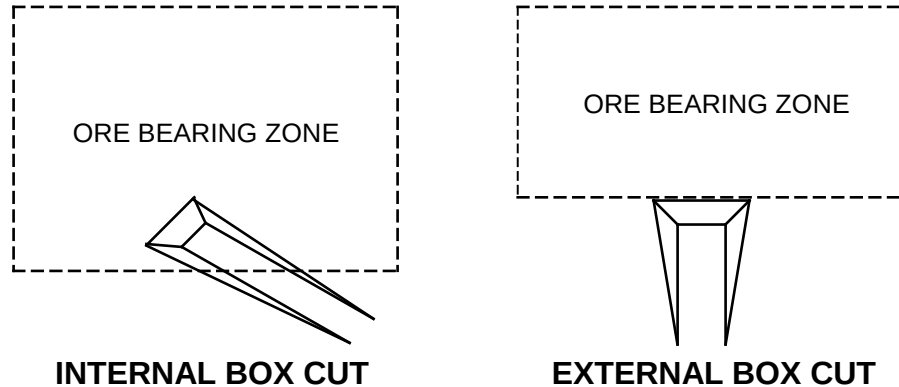
Box cuts are of two types – Internal box cut and External box cut.

Internal box cut

When the box cut is located fully or partially on the mineralized zone, it is called an internal box cut. This is applicable for all types of deposit. The cut follows a direction that is usually oblique to both the strike and dip direction. Generally, the direction is so chosen that the haul road ramp formed by this cut and subsequent cuts will not have unnecessary steep turning at any position.

External box cut

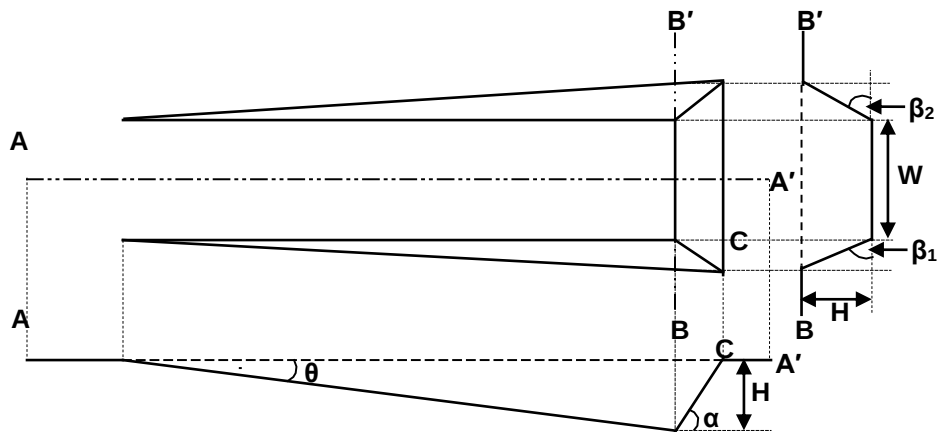
When the box cut is placed totally outside the mineralized zone, it is called an external box cut. This is applicable only for shallow and gently dipping bedded deposits. The cut is generally located at the middle of the rise-most side.



2.1.2. Box Cut - Parameters and Methods

Parameters

The main parameters of a box cut are – Maximum level difference (H), width of the floor (W), longitudinal inclination of the floor (θ), front slope angle (α) and side slope angles (β_1 & β_2).



BOX CUT PARAMETERS

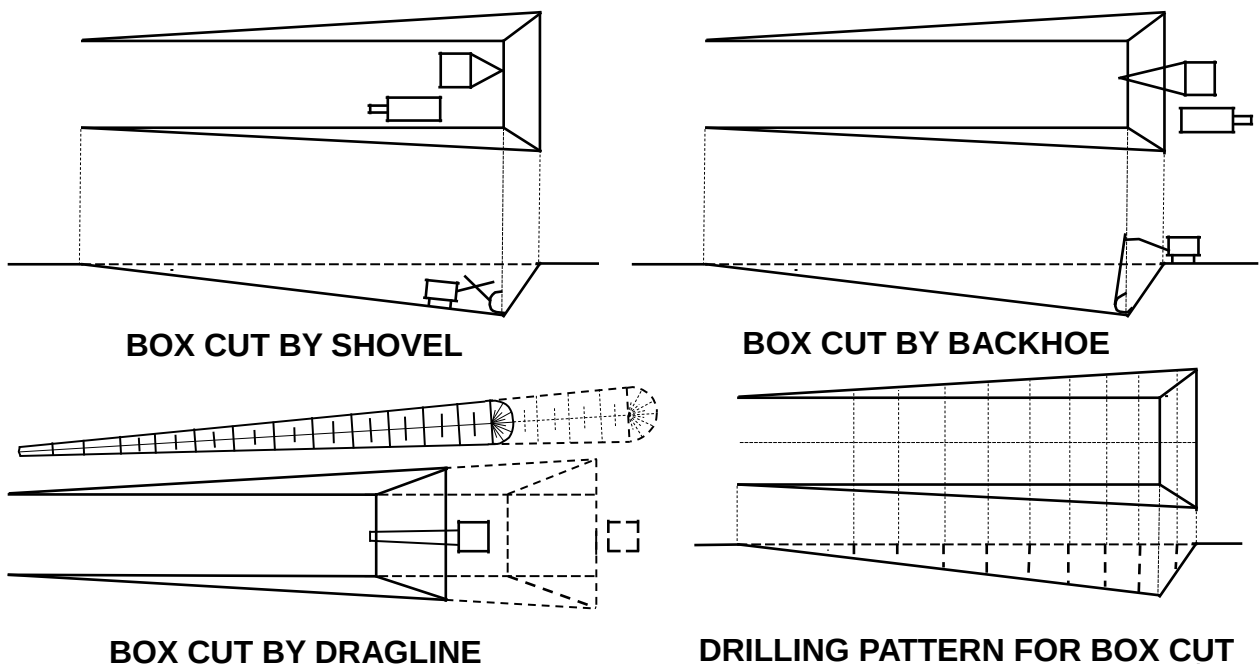
Maximum level difference (H) or the height of the box cut is generally equal to the proposed bench height. **Width of the floor** (W) of the box cut should be such that the movement of the machinery is smooth. If shovel – dumper combination is used for the box cut, then there should be sufficient space for the shovel movement for excavation & loading and the width should be at least three times the width of the dumper being used. **Longitudinal inclination of the floor** (θ) of the box cut depends on the intended use of it at later dates. If it is planned that the box cut will form a part of the haul road ramp, then the cut is generally made with an inclination between 4.086 deg. (1 in 14) and 4.764 deg. (1 in 12). **Front slope angle** (α) and **side slope angles** (β_1 & β_2) are generally dependent on the rock-mass properties (physico-mechanical) and the pore pressure condition at the cut site, and the maximum height of the cut. The front slope angle is generally

steeper than the side slope angles. The side slope angles are generally same, but under certain conditions they may be different.

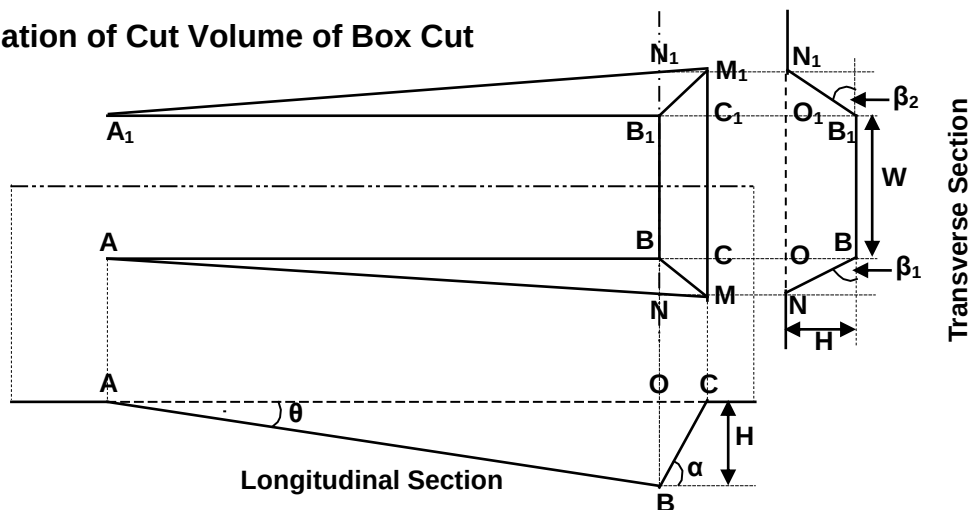
Methods

For establishing box cut shovel – dumper combination or backhoe – dumper combination is generally used. However, in certain cases dragline can also be used for the purpose. The schemes for establishing box cut by shovel – dumper combination, backhoe – dumper combination and dragline are shown below.

For establishing box cut in some situations (in hard rocks), it may be necessary to adopt drilling & blasting to loosen the rock mass to facilitate its excavation & loading. For this, holes of different depths are required to be drilled and blasted to form the inclined floor of the cut.



2.1.3. Calculation of Cut Volume of Box Cut



The points O and O₁ are the projections of points B and B₁ respectively on the surface

LECTURE NOTES – SURFACE MINING

Considering the longitudinal section and the transverse section,

$OB = O_1B_1 = H$ (height of the cut)

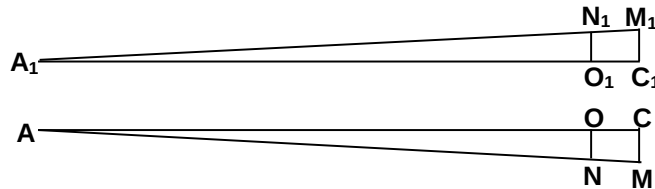
From $\triangle ABO$, $\tan\theta = OB / AO \Rightarrow AO = OB / \tan\theta = H / \tan\theta$

From $\triangle OBC$, $\tan\alpha = OB / OC \Rightarrow OC = OB / \tan\alpha = H / \tan\alpha$

$AC = AO + OC \Rightarrow AC = H [(1 / \tan\theta) + (1 / \tan\alpha)] \Rightarrow AC = H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)]$

From $\triangle OBN$, $\tan\beta_1 = OB / ON \Rightarrow ON = OB / \tan\beta_1 = H / \tan\beta_1$

From $\triangle O_1B_1N_1$, $\tan\beta_2 = O_1B_1 / O_1N_1 \Rightarrow O_1N_1 = O_1B_1 / \tan\beta_2 = H / \tan\beta_2$



Again considering the surface projections of surfaces $ABCMNA$ and $A_1B_1C_1M_1N_1A_1$,

From $\triangle AON$ and $\triangle ACM$, $MC / ON = AC / AO = (AO + OC) / AO = 1 + (OC / AO)$

$\Rightarrow MC = ON \times [1 + (OC / AO)]$

$\Rightarrow MC = (H / \tan\beta_1) \times [1 + (H / \tan\alpha) / (H / \tan\theta)] = (H / \tan\beta_1) \times [1 + (\tan\theta / \tan\alpha)]$

$\Rightarrow MC = (H / \tan\beta_1) \times [(\tan\alpha + \tan\theta) / \tan\alpha] = H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\beta_1)]$

Similarly from $\triangle A_1O_1N_1$ and $\triangle A_1C_1M_1$,

$M_1C_1 / O_1N_1 = A_1C_1 / A_1O_1 = (A_1O_1 + O_1C_1) / A_1O_1 = 1 + (O_1C_1 / A_1O_1)$

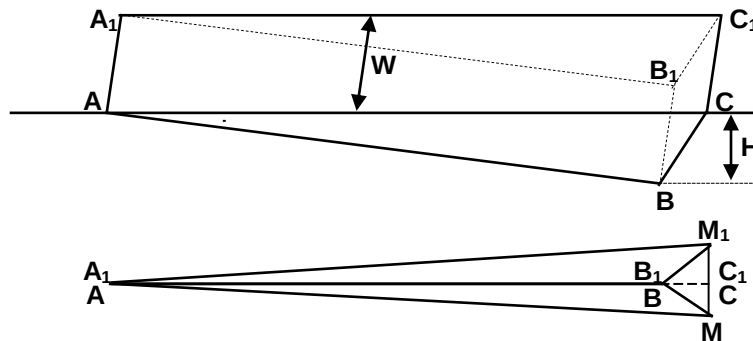
$\Rightarrow M_1C_1 = O_1N_1 \times [1 + (O_1C_1 / A_1O_1)] \Rightarrow M_1C_1 = H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\beta_2)]$

$\Rightarrow MM_1 = MC + M_1C_1 = H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\beta_1)] + H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\beta_2)]$

$\Rightarrow MM_1 = H \times [(\tan\alpha + \tan\theta) / \tan\alpha] \times [(1 / \tan\beta_1) + (1 / \tan\beta_2)]$

If $\beta_1 = \beta_2 = \beta$, then $MM_1 = 2H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\beta)]$

For calculation of the cut volume, the whole volume has been divided into three segments – segment $ABCC_1B_1A_1$, segment $ABCMN$ and segment $A_1B_1C_1M_1N_1$. Of these, the segment $ABCC_1B_1A_1$ is an inverted triangular prism of width W , and when the other two segments $ABCMN$ and $A_1B_1C_1M_1N_1$ are joined together they form an inverted pyramid with a triangular base AMM_1A_1 as shown below.



Let,

V_1 be the volume of the prism $ABCC_1B_1A_1$ and V_2 be the volume of the pyramid MM_1BA .

So, the total cut volume, $V = V_1 + V_2$

$$V_1 = W \times \text{Area of } \triangle ABC = W \times (1/2) \times AC \times OB = (W/2) \times [H \times (\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)] \times H$$

$$\Rightarrow V_1 = H^2 \times (W/2) \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)]$$

$$V_2 = (1/3) \times H \times \text{Area of } \triangle AMM_1 = (1/3) \times H \times (1/2) \times (MC + M_1C_1) \times AC$$

$$\Rightarrow V_2 = (H/6) \times H \times [(\tan\alpha + \tan\theta) / \tan\alpha] \times [(1/\tan\beta_1) + (1/\tan\beta_2)] \times H \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)]$$

$$\Rightarrow V_2 = H^2 \times (H/6) \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)] [(\tan\alpha + \tan\theta) (\tan\beta_1 + \tan\beta_2) / (\tan\alpha \tan\beta_1 \tan\beta_2)]$$

$$\text{If } \beta_1 = \beta_2 = \beta, V_2 = H^2 \times (H/3) \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)] [(\tan\alpha + \tan\theta) (\tan\beta) / (\tan\alpha \tan\beta)]$$

$$V = V_1 + V_2 = H^2 \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)] [(W/2) + (H/6) \times (\tan\alpha + \tan\theta) (\tan\beta_1 + \tan\beta_2) / (\tan\alpha \tan\beta_1 \tan\beta_2)]$$

$$\text{If } \beta_1 = \beta_2 = \beta, V = H^2 \times [(\tan\alpha + \tan\theta) / (\tan\alpha \tan\theta)] [(W/2) + (H/3) \times (\tan\alpha + \tan\theta) / (\tan\alpha \tan\beta)]$$

2.1.4. Factors affecting the Selection of Box-Cut Site

Selection of location of box cut for any surface mine is of utmost importance as it has far reaching effects on the economy of the mine. The location should be so chosen that there is a minimum of foreseeable trouble with uncovering the ore body and a maximum of economy. In selecting the optimum box cut location, the following factors are necessary to be taken into consideration.

Site accessibility: The site for establishing the box cut should be easily accessible/reachable for men and machinery.

Minimum excavation requirement: The location of box cut should be so chosen that the material to be excavated to reach the ore body will be as less as possible. This will ensure early production of ore from the mine, and this is very important from the view point of the economics of mine operation.

Availability of dumping space: The cut material should preferably be handled only once and stacked off the mineral bearing zone. For dumping the excavated material, sufficient space should be available nearby. This will improve the economy of transport of the cut material.

Pit haulage plan: If the box cut is planned to be a part of the pit haulage system, it must match with the haulage plan for the complete pit. The roads, as far as practicable, should be put in but once and used as much as possible.

Overall mine plan: The box cut location should be so chosen that it matches with the overall mine plan including the waste dump locations, sites for infrastructural facilities and sites for other facilities. It should not hinder the pit progress/expansion at a later date. Also the location should be such that the overall economy of ore and waste transport over the whole mine life is maintained.

Water condition: Drainage is one of the most important aspects for the box cut. Surface drainage must be diverted, preferably by gravity. If possible, the location of the cut should be so chosen that the problems of accumulation of water from surface water and/or ground water are not there or minimum.

Geological disturbance: The box cut location should preferably be free of any geological discontinuities and structural disturbances.

Reclamation requirement: The reclamation requirements may dictate that the box cut material should be dumped/emplaced at particular location. This may affect the location of the cut in order to satisfy such requirements most economically.

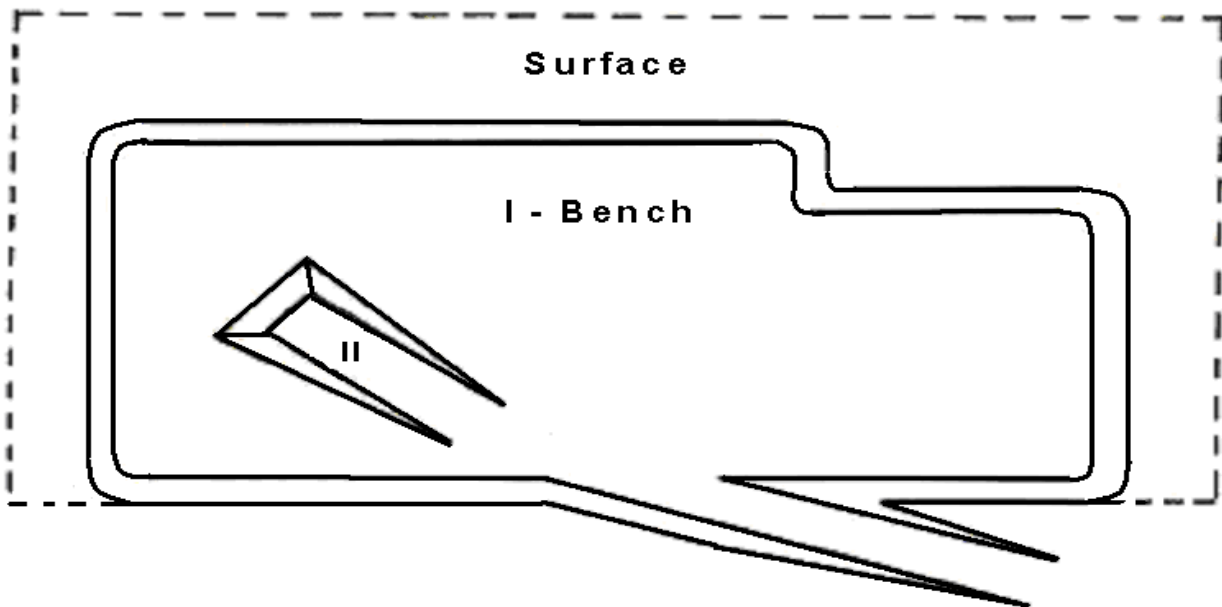
If the box cut is a part of the pit haul road system, particularly in case of external box cut for a bedded deposit, preferably the cut should be placed at the middle of the rise-most side of the deposit for the economy of transportation provided that the location is not dictated otherwise by some other factors.

2.2. Production Benches

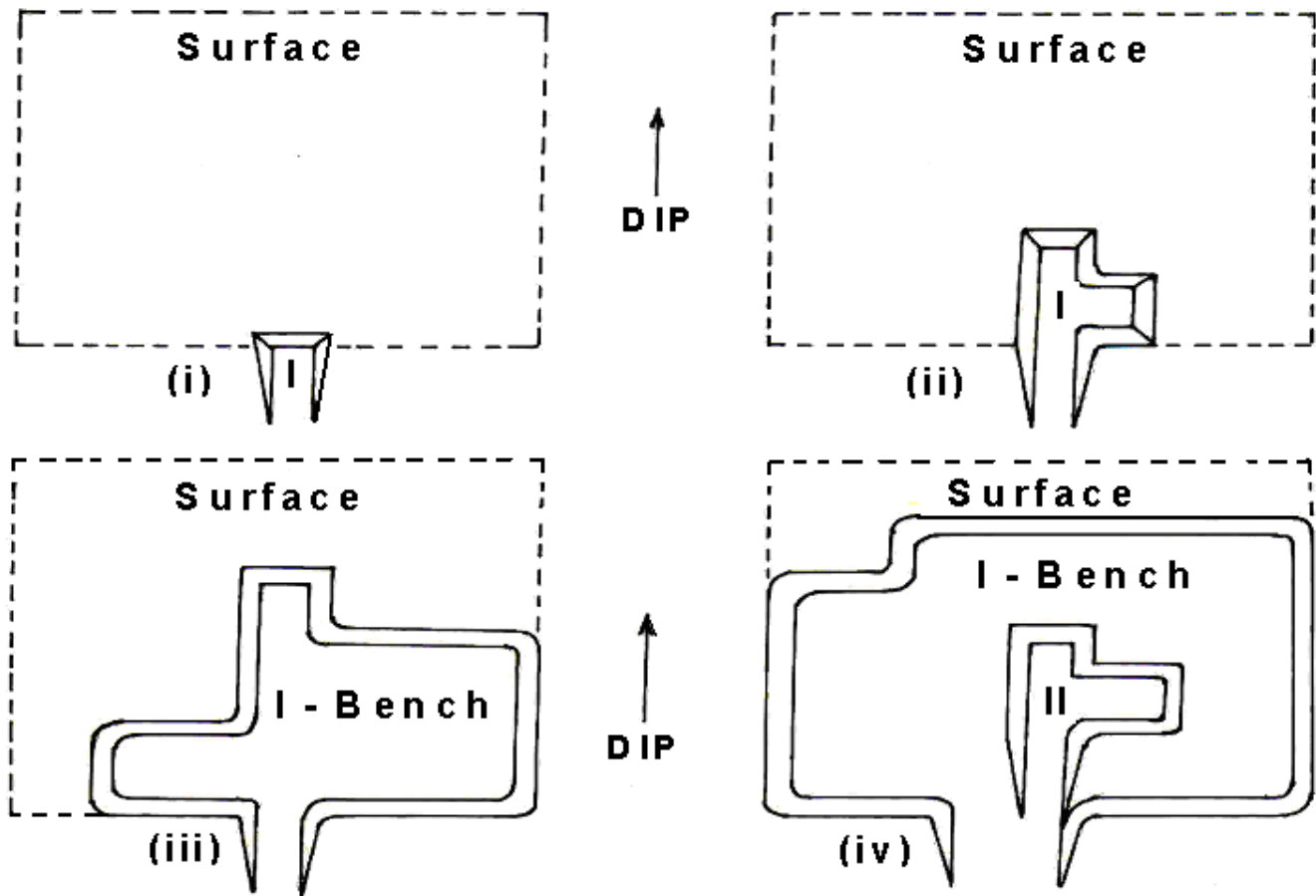
2.2.1. Formation of Benches

Once the box cut is established upto the 1st bench floor level, the development of the 1st bench is started. Once the 1st bench is developed over a considerable area, the development of subsequent benches are done by two methods – by establishing additional cuts or by extending the existing box cut after widening the same.

Formation of benches by establishing additional cut



In this method, the first bench (I – Bench) is developed by extending the initial box-cut. Proper working face is established and the bench of designed height and slope is advanced sufficiently. This is followed by establishing the next cut (second cut), a separate one from the first cut, leaving sufficient gap/space between the two cuts to allow unrestricted movement of dumpers between the first bench and the surface. The second cut is generally established oblique to the first cut. Once the second cut is established fully, the development of the second bench of designed height and slope starts; and the same process continues for development of subsequent benches.

Formation of benches by extending the existing box cut

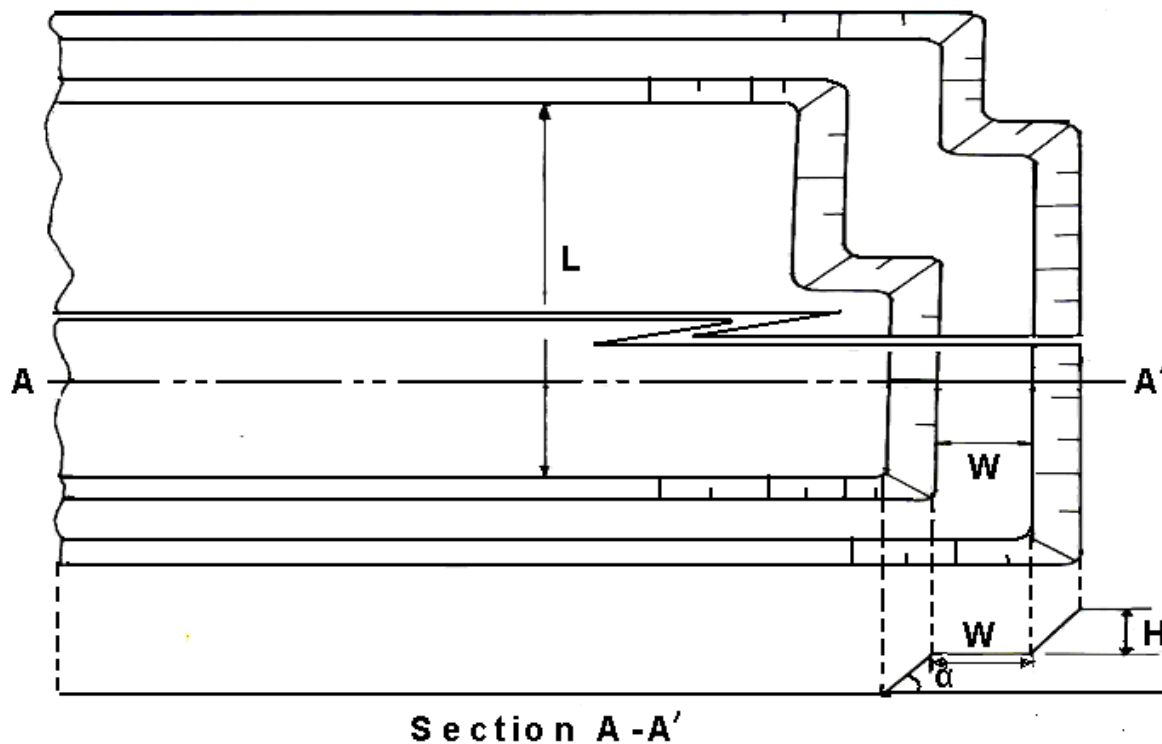
In this case, the first bench (I – Bench) is developed by extending the initial box-cut. Proper working face is established and the bench of designed height and slope is advanced sufficiently (figures i to iii). This is followed by establishing the next cut (second cut) by extending the first cut after widening the same so as to leave sufficient gap/space to allow unrestricted movement of dumpers between the first bench and the surface. The second cut effectively becomes as an extension of the first cut. Once the second cut is established fully, the development of the second bench of designed height and slope starts; and the same process continues for development of subsequent benches.

2.2.2. Bench Parameters and their selection

A surface mine bench is characterized by the following bench parameters – Bench height, Bench width, Bench length, Bench face slope and Bench floor slope.

Bench height is the vertical distance between the crest and toe of a bench. As specified by the Directorate General of Mines Safety (DGMS), it shall not be more than the -

- (i) maximum digging height of the excavator excavating the bench in case of hard formation
- (ii) maximum digging height of the excavator excavating the bench + 3m in case of soft formation



H - Bench Height W – Bench Width L – Bench Length α - Bench Face Slope

BENCH PARAMETERS

From the operational point of view, the bench height is decided based on the following criteria – optimum digging height of the excavator for efficient excavation, one pass drilling length should preferably be more than the bench height, and the blending requirement dictated by the geological parameters.

For benches worked by rope shovels and draglines, the bench height is generally selected on the basis of optimum digging height of the excavator. In case of hydraulic excavators, the bench height is generally selected on the basis of maximum digging height. In case of bucket wheel excavator (BWE) workings, the bench height depends on – the height of the cutting boom pivot point, the effective length of cutting boom including the cutting wheel and the allowable maximum angle of inclination of cutting boom in the vertical plane limited by the dynamic angle of repose of the cut material.

Bench width is the horizontal distance between the crest of a bench and the toe of the immediately upper bench. As specified by the DGMS, it shall be more than -

- (i) the bench height
- (ii) the maximum width of the largest machine + 2m
- (iii) three times the width of largest size dumper that is plying on the bench

From the operational point of view, minimum width of a working bench should be equal to $0.8 \times \text{bench height} + \text{dumper width} + \text{shovel dimension} + 4\text{m}$.

Bench length in a surface mine is the length (straight or curve as the case may be) measured along the floor of the bench between its two extremities. It is generally decided based on the operational requirements and conveniences. The minimum bench length required is generally dictated by the frequency of primary blasting and the advancement of the face along the length of the bench per blast, and also on the type of excavator used.

In a mine that is being worked by shovel – dumper combination with full scale blasting, minimum bench length required may be calculated as follows.

Let,

- (i) The frequency of primary blasting in the bench is once per week
- (ii) The volume of material to be handled per week from the bench is V
- (iii) The height and width of the bench are H and W respectively

The length of broken material required per week, $I = V / (H \times W)$

The minimum bench length, $L = 3 \times I$

The minimum bench length, L shall be equal to $3 \times I$ as an additional length of I will be required for drilling while the blasted material is being handled by the excavator and another extra length of I for maintaining the flexibility.

For dragline benches, the minimum bench length should be 300m to 500m in view of marching requirements and operational conveniences.

For the mines being worked by bucket wheel excavator (BWE) – conveyor combination, the minimum bench length should be decided on the basis of the production requirement, the method of work of BWE (full block or half block), shiftable belt conveyor shifting frequency and other operational conveniences.

The maximum bench length is generally selected on the basis of depositional features and/or transport requirement.

Bench face slope is the slope of a line (in the vertical plane) that joins the toe and the crest of the bench. It is generally selected considering the requirements of slope stability and inclined hole blasting, and criterion of complete removal of blasted material by the excavator excavating the bench.

Bench floor slope: Generally 1% to 2% longitudinal slope is provided on the bench floor to facilitate the drainage of water to the sump area.